Deakin University
Faculty of Science, Engineering and Built Environment

School of IT

Assignment 1 Report

SIT215 – Computational Intelligence

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Tasks Completed	Completed (✓)	Notes (Optional)
Task 1: Environment modelling and problem formulation	☐	
Task 2: Search-based agent implementation and comparison • A^* • Selected uninformed algorithm	☐	
Task 3: Success metrics and evaluation	☐	
Task 4: New heuristic design and performance	☐	

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1. Problem Formulation

1.1 Problem Definition

1.2 Environment Description

1.3 Map and Graph Representation

1.4 Constraints in the Problem

1.5 Goal State and Feasibility

2. Modelling Approach

2.1 Environment Abstraction into State Space

2.2 State Representation

2.3 Successor Function

2.4 Cost Model

2.5 PEAS Framework

Component	**Description**
Performance Measure	
Environment	
Actuators	
Sensors	

Table 1: PEAS framework for the truck delivery agent.

2.6 Relaxed Version of the Problem

2.7 Justification of Modelling Choices

3. Search Strategies and Heuristic Design

3.1 Benchmark Algorithm: A*

3.2 Baseline Heuristic: Manhattan Distance

3.3 Selected Uninformed Search Algorithm

3.4 Comparison Rationale

3.5 Heuristic Properties

3.6 New Heuristic Design

3.7 Expected Impact of the New Heuristic

4. Implementation Overview

4.1 Overall System Structure

4.2 Data Structures

4.3 Node Expansion Logic

4.4 Constraint Validation Logic

4.5 Implementation of A*

4.6 Implementation of the Uninformed Search Algorithm

4.7 Implementation of the New Heuristic

4.8 Code Snippet References

Listing 1: Placeholder for annotated code snippet with line numbers.

```
# Paste relevant code snippet here
# Make sure line numbers are visible in the PDF output
```

Listing 2: A* Search Algorithm

```
def astar(graph, start, goal, heuristic):
    open_set = PriorityQueue()
    open_set.put((0, start))

    came_from = {}
    g_score = {start: 0}

    while not open_set.empty():
        _, current = open_set.get()

        if current == goal:
            return reconstruct_path(came_from, current)

        for neighbor, cost in graph[current]:
            tentative = g_score[current] + cost

            if neighbor not in g_score or tentative < g_score[neighbor]:
                came_from[neighbor] = current
                g_score[neighbor] = tentative
                f_score = tentative + heuristic(neighbor, goal)
                open_set.put((f_score, neighbor))
```

5. Results and Evaluations

5.1 Experimental Setup

5.2 Test Cases

Case	Start Node	Goal Node	Notes
1			
2			
3			

Table 2: Start-goal test cases used for evaluation.

5.3 Performance Metrics

5.4 Comparison of A* and Uninformed Search

Case	Algorithm	Path Found	Path Cost	Nodes Expanded	Runtime	Optimal?
1	A*					
1	Uninformed					
2	A*					
2	Uninformed					
3	A*					
3	Uninformed					

Table 3: Performance comparison of A* and the selected uninformed search algorithm.

5.5 Evaluation of Intelligence Quality

5.6 Comparison of Heuristics

Case	Heuristic	Path Cost	Nodes Expanded	Runtime	Comments
1	Manhattan				
1	New heuristic				
2	Manhattan				
2	New heuristic				
3	Manhattan				
3	New heuristic				

Table 4: Comparison between Manhattan distance and the new heuristic.

5.7 Visualisations

Figure 1: Placeholder for result visualisation (e.g., runtime or node expansion comparison).

5.8 Discussion

6. Conclusion

Acknowledgement

References